

# RIYA

AI Engineer

+91 8570075840 | [riya.rk006@gmail.com](mailto:riya.rk006@gmail.com) | [linkedin.com/in/riyark006](https://www.linkedin.com/in/riyark006) | [github.com/riya-py](https://github.com/riya-py)

---

## SUMMARY

Computer Science undergraduate and aspiring AI Engineer with hands-on experience building applied machine learning and GenAI systems in Python, including retrieval-augmented generation (RAG) pipelines, vector search engines, and LLM-orchestrated applications using LangChain and LangGraph. Comfortable across the ML lifecycle: data processing, model training and evaluation, embedding-based retrieval, and deploying models behind REST APIs. Strong foundation in data structures and algorithms and backend engineering, with a track record of independently building, documenting, and open-sourcing complete AI projects from scratch. A fast learner eager to contribute to production-grade AI/ML systems as part of an engineering team.

---

## TECHNICAL SKILLS

**AI / ML & GenAI:** RAG (Retrieval-Augmented Generation), LangChain, LangGraph, TensorFlow, PyTorch, Keras, Scikit-learn, XGBoost, OpenCV, BERT

**Vector Search & Databases:** HNSW, KD-Tree, Vector Databases, SQL

**Languages:** Python, Java, C, C++, SQL

**Data & Visualization:** Pandas, NumPy, Matplotlib, Seaborn, Tableau, Plotly, Power BI

**Tools & Platforms:** Git, GitHub, Linux, Docker, Streamlit, Ollama, REST APIs

**Core CS:** Data Structures & Algorithms, REST API Design, Multi-threaded Systems, OOP, System Design

**Soft Skills:** Analytical Thinking, Collaboration, Adaptability, Problem Solving, Fast Learning

---

## PROJECTS

### VectorDB — Vector Database & RAG Pipeline from Scratch

2026

*Tech Stack:* Python, HNSW, KD-Tree, REST API, LangChain, Ollama, HTML/CSS/JS

- Built a vector database from scratch in Python, implementing and benchmarking three nearest-neighbor search algorithms (HNSW, KD-Tree, Brute Force) across three distance metrics (Cosine, Euclidean, Manhattan).
- Implemented the HNSW (Hierarchical Navigable Small World) multilayer graph algorithm to achieve approximate  $O(\log N)$  semantic search, the same core algorithm used by production vector databases such as Pinecone and Weaviate.
- Engineered a Retrieval-Augmented Generation (RAG) pipeline using LangChain that chunks and embeds user documents locally via Ollama, retrieves the most relevant chunks through the HNSW index, and generates grounded answers using a local LLM.
- Exposed a full REST API and built a browser-based UI with a live 2D PCA visualization of the vector space.

**GitHub:** [github.com/riya-py/OWN-AI](https://github.com/riya-py/OWN-AI)

### DPI Engine — Multi-threaded Network Packet Analyzer

2026

*Tech Stack:* Python, Multi-threading, TCP/IP, TLS, PCAP, Scapy

- Built a deep packet inspection engine in Python that parses raw PCAP network captures at the Ethernet, IP, and TCP layers and extracts TLS Client Hello SNI fields to identify application traffic (e.g., YouTube, Facebook) even within encrypted HTTPS connections.
- Designed a multi-threaded producer-consumer pipeline (reader, load-balancer, and fast-path worker threads) connected by thread-safe queues, using consistent hashing on the connection 5-tuple so all packets of a flow are processed by the same worker.
- Implemented configurable, flow-level traffic-blocking rules by source IP, application type, and domain name, with real-time classification reporting.

**GitHub:** [github.com/riya-py/Packet-analyzer](https://github.com/riya-py/Packet-analyzer)

---

## EDUCATION

**B.Tech in Computer Science — TIT&S, Bhiwani, Haryana**

Expected: Summer 2027

---

## CERTIFICATIONS & ACHIEVEMENTS

- Certificate of Appreciation — Secured a position in the Top 40 teams (Team Neural Network) at Hackfluence 2026, a hackathon organized by CodeBenders, IGDTUW in collaboration with Dropp.